

ATPL EASA STUDY GUIDE | 033 - FLIGHT PLANNING

Chapter 3: Critical Points: Point of No Return (PNR) & Equi-Time Point (ETP)

Syllabus Source: CAE Oxford (pp. 83-120) | Pass Target: >= 90% AviationExam

1. POINT OF NO RETURN (PNR / POINT OF SAFE RETURN)

The Point of No Return (PNR) is the furthest point along track to which an aircraft can fly and still return to the departure aerodrome (or en-route base) with mandatory reserves intact:

- PNR Concept: Beyond the PNR, the aircraft CANNOT RETURN to base; it is committed to flying forward. PNR depends exclusively on: (1) Safe Endurance (E), (2) Groundspeed Out (GS_out), and (3) Groundspeed Home (GS_home).
- Safe Endurance (E): Total fuel on board MINUS (Alternate Fuel + Final Reserve Fuel + Taxi Fuel). Only usable cruise fuel can be spent flying to PNR and back.
- The PNR Master Formula: Time to PNR (T_{out}) = (Safe Endurance (E) x GS_home) / (GS_out + GS_home)
- Distance to PNR: Distance_PNR = Time to PNR x GS_out

2. EQUI-TIME POINT (ETP / CRITICAL POINT)

The Equi-Time Point (ETP) is the point along track from which it takes EQUAL FLYING TIME to continue to destination or to return to departure:

- ETP Concept: In zero wind, ETP is exactly half-way (50% distance). BUT WITH WIND, ETP MOVES INTO THE WIND! (A headwind on the way out shifts ETP closer to destination; a tailwind shifts ETP closer to departure).
- The ETP Master Formula: Distance to ETP (D_{ETP}) = (Total Leg Distance (D) x GS_home) / (GS_out + GS_home)
- Time to ETP: Time to ETP = D_{ETP} / GS_{out}

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Critical Point (ETP) & Safe Return**

Problem: Total Distance D = 1,200 NM. TAS = 400 kt. Headwind Outbound = 50 kt. Safe Endurance E = 4.0 hr. Step 1: Outbound GS (O) = 400 - 50 = 350 kt. Homebound GS (H) = 400 + 50 = 450 kt. Step 2: Distance to Equi-Time Point (ETP): - Formula: $Dist_{ETP} = (D \times H) / (O + H) = (1,200 \times 450) / (350 + 450) = 540,000 / 800 = 675$ NM. Step 3: Time to Point of Safe Return (PNR): - Formula: $Time_{PNR} = (E \times H) / (O + H) = (4.0 \times 450) / (350 + 450) = 1,800 / 800 = 2.25$ hr (2 hr 15 min). Step 4: $Dist_{PNR} = 2.25$ hr x 350 kt = 787.5 NM from departure.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Finding PNR and ETP with Wind

SCENARIO (Classic Exam Problem): An oceanic flight is planned between Airport A and Airport B: - Total Distance (D) = 1,200 NM - Aircraft TAS = 400 knots - En-route Wind: 50 knots direct Headwind on outbound leg (Tailwind returning home!) - Total Safe Endurance (E) = 4.0 hours QUESTION: Calculate (1) Distance to ETP from Airport A, and (2) Time and Distance to PNR. SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate Outbound and Inbound Groundspeeds: - $GS_{out} = TAS - Headwind = 400 \text{ kt} - 50 \text{ kt} = 350 \text{ knots}$. - $GS_{home} = TAS + Tailwind = 400 \text{ kt} + 50 \text{ kt} = 450 \text{ knots}$. - $GS_{sum} = GS_{out} + GS_{home} = 350 + 450 = 800 \text{ knots}$. Step 2: Calculate Distance to ETP (Equi-Time Point): - Formula: $D_{ETP} = (Total \text{ Distance} \times GS_{home}) / (GS_{out} + GS_{home})$ - $D_{ETP} = (1,200 \text{ NM} \times 450 \text{ kt}) / 800 \text{ kt} = 540,000 / 800 = 675 \text{ NM}$ from Airport A! - (Notice: Because of the headwind, ETP shifted forward from 600 NM to 675 NM!) Step 3: Calculate Time to PNR (Point of No Return): - Safe Endurance (E) = 4.0 hours. - Formula: $T_{out} = (E \times GS_{home}) / (GS_{out} + GS_{home})$ - $T_{out} = (4.0 \text{ hrs} \times 450 \text{ kt}) / 800 \text{ kt} = 1,800 / 800 = 2.25 \text{ hours}$ (2 hrs 15 min)! Step 4: Calculate Distance to PNR: - $Distance_{PNR} = T_{out} \times GS_{out} = 2.25 \text{ hrs} \times 350 \text{ kt} = 787.5 \text{ NM}$ from Airport A! FINAL ANSWER: Distance to ETP = 675 NM. Time to PNR = 2.25 hours (787.5 NM).

[!] EXAM TRAP (AVIATIONEXAM) - The ETP Moves Into the Wind Trap

- If you have a HEADWIND outbound: GS_{home} is FASTER than GS_{out} . Therefore, ETP shifts FORWARD (closer to destination)! - If you have a TAILWIND outbound: GS_{home} is SLOWER. Therefore, ETP shifts BACKWARD (closer to departure)! - Always remember: In the formula numerator, it is ALWAYS GS_{home} ! ($D_{ETP} = D \times GS_{home} / Sum_{GS}$).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In the formula for calculating distance to the Equi-Time Point (ETP), which groundspeed appears in the NUMERATOR?

- [A]: GS outbound
- [B]: GS HOME (return groundspeed)
- [C]: Average TAS
- [D]: Crosswind component

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The formula is: $Distance \text{ to ETP} = (Total \text{ Distance} \times GS_{home}) / (GS_{out} + GS_{home})$. GS_{home} is in the numerator.

Q2: On an oceanic flight with a strong HEADWIND on the outbound leg, where will the Equi-Time Point (ETP) be located?

- [A]: Exactly at the midpoint of the route
- [B]: Displaced TOWARDS THE DESTINATION (into the wind)
- [C]: Displaced towards departure
- [D]: ETP ceases to exist

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Because returning home has a tailwind (faster GS), you can fly further towards destination before reaching the point where flying forward takes the same time as flying back.

Q3: What single factor determines the location of the POINT OF NO RETURN (PNR) for a given aircraft and wind?

- [A]: Only aircraft mass
- [B]: The SAFE ENDURANCE of the aircraft (usable fuel available to return)
- [C]: Runway length
- [D]: Flap angle

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. PNR is purely fuel-limited. It is the furthest point from which the aircraft can return with mandatory reserve fuel intact: $T = (E \times GS_{home}) / (GS_{out} + GS_{home})$.